

U.S. Preventive Services Task Force (USPSTF) Recommendation Statement

Title

Screening for Osteoporosis to Prevent Fractures: US Preventive Services Task Force Recommendation Statement

Source

US Preventive Services Task Force. JAMA. Published January 14, 2025.

1.0 Recommendation Summary

Population: Women 65 years or older

Recommendation: The USPSTF recommends screening for osteoporosis with bone measurement testing to prevent osteoporotic fractures in women 65 years or older.

Grade: Grade B

Population: Postmenopausal women younger than 65 years at increased risk of osteoporosis

Recommendation: The USPSTF recommends screening for osteoporosis with bone measurement testing to prevent osteoporotic fractures in postmenopausal women younger than 65 years who are at increased risk of osteoporosis, as determined by a formal clinical risk assessment tool.

Grade: Grade B

Population: Men

Recommendation: The USPSTF concludes that the current evidence is insufficient to assess the balance of benefits and harms of screening for osteoporosis to prevent osteoporotic fractures in men.

Grade: I statement

2.0 Clinical Considerations and Target Population

These recommendations apply to adults 40 years or older without known osteoporosis or history of fragility fractures. They do not apply to persons with secondary osteoporosis due to an underlying medical condition (eg, hyperparathyroidism, malabsorption, hypogonadism) or chronic use of medications associated with bone loss (eg, long-term glucocorticoids).

Potential Preventable Burden: Approximately 10 million Americans have osteoporosis and another 43 million have low bone mass (osteopenia). Fractures of the hip, vertebrae, and wrist are associated with chronic pain, disability, loss of independence, and increased mortality.

2.1 Risk Assessment in Postmenopausal Women Younger Than 65 Years

For postmenopausal women younger than 65 years, clinicians should first evaluate clinical risk factors for fracture. Risk factors include advancing age, low body weight (BMI < 21 kg/m²), parental history of hip fracture, current smoking, high alcohol intake (≥3 drinks/day), and history of falls.

Formal clinical risk assessment tools such as FRAX (Fracture Risk Assessment Tool without BMD), the Simple Calculated Osteoporosis Risk Estimation (SCORE), the Osteoporosis Risk Assessment Instrument (ORAI), and the Osteoporosis Self-Assessment Tool (OST) can be used to estimate fracture risk. Women whose 10-year major osteoporotic fracture risk exceeds that of a 65-year-old White woman without additional risk factors (typically ≥8.4% by FRAX) should undergo dual-energy x-ray absorptiometry (DXA) screening.

Grade: Grade B

USPSTF Osteoporosis Screening: Screening Tests & Intervals

3.0 Screening Tests (Bone Measurement Testing)

Central dual-energy x-ray absorptiometry (DXA) of the hip and lumbar spine is the standard diagnostic and screening tool for measuring bone mineral density (BMD) and predicting fracture risk.

The World Health Organization (WHO) defines osteoporosis as a T-score of -2.5 or lower at the femoral neck or lumbar spine compared with a young adult reference population. Osteopenia is defined as a T-score between -1.0 and -2.5.

Peripheral DXA (calcaneus, forearm), quantitative ultrasound (QUS), and quantitative computed tomography (QCT) predict fractures but lack standardized treatment thresholds comparable to central DXA.

3.1 Screening Intervals and Rescreening

Current evidence is insufficient to determine the optimal interval for rescreening individuals whose initial screening test is normal. A minimum interval of 2 or more years may be needed to detect a statistically significant change in BMD.

For women with normal initial BMD (T-score > -1.0), studies suggest that rescreening intervals can be extended (up to 10 to 15 years for women with normal bone density or mild osteopenia), whereas women with baseline T-scores close to -2.5 may warrant earlier rescreening.

4.0 Suggestions for Practice Regarding the I Statement in Men

The USPSTF concludes that evidence on the benefits and harms of routine screening for osteoporosis in men is insufficient (Grade I statement).

In deciding whether to screen men for osteoporosis, clinicians may consider the presence of specific risk factors such as previous fragility fracture, hypogonadism, chronic glucocorticoid therapy, or androgen deprivation therapy. In the absence of definitive trials showing that screening reduces fracture rates in asymptomatic men, routine population screening cannot be recommended universally.

Grade: I statement

5.0 Treatment and Interventions

Pharmacotherapy with bisphosphonates (alendronate, risedronate, zoledronic acid), denosumab, selective estrogen receptor modulators (raloxifene), or anabolic agents (teriparatide, abaloparatide) has been proven in randomized trials to reduce vertebral and nonvertebral fractures in women with osteoporosis.

Adequate intake of calcium (1000-1200 mg daily) and vitamin D (600-800 IU daily), regular weight-bearing exercise, fall prevention strategies, and smoking cessation are recommended baseline measures for bone health.

Grade: Grade B